

随过 作业3 14.17.21.23

14.

(1) $I = \{0, 1, \dots, N\}$

$$P_{ii} = (1-p) \frac{N-i}{N} + p \frac{i}{N} = \frac{(1-p)(N-i) + pi}{N}$$

$$P_{i,i+1} = p \frac{N-i}{N}$$

$$P_{i,i-1} = (1-p) \frac{i}{N}$$

(2)

$$P = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

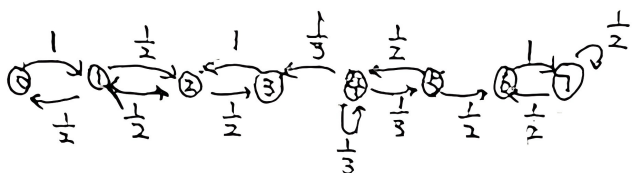
$$P = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 3 & 0 & 1 & 0 \end{bmatrix}$$

$$\pi = \pi P$$

$$\begin{cases} \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \\ \pi_0 = \frac{1}{2}\pi_0 + \frac{1}{2}\pi_1 \\ \pi_1 = \frac{1}{2}\pi_0 + \frac{1}{2}\pi_1 + \frac{1}{2}\pi_2 \\ \pi_2 = \frac{1}{2}\pi_1 + \frac{1}{2}\pi_2 \end{cases} \Rightarrow \begin{cases} \pi_0 = \frac{1}{8} \\ \pi_1 = \frac{3}{8} \\ \pi_2 = \frac{4}{8} \\ \pi_3 = \frac{1}{8} \end{cases}$$

$$\mu_0 = \frac{1}{\pi_0} = 8$$

17.



(1)

$\{0, 1, 2, 3\}$ 闭

$\{4, 5\}$ 不闭

$\{6, 7\}$ 闭

(2)

0, 1, 2, 3 正常返 $d=2$

4, 5 暂留

6, 7 正常返 $d=1$

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 1 \\ 3 & 0 & 0 & 1 \end{bmatrix} \begin{cases} \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \\ \pi_0 = \frac{1}{2}\pi_1 \\ \pi_1 = \pi_2 + \frac{1}{2}\pi_0 \\ \pi_2 = \frac{1}{2}\pi_1 \end{cases} \Rightarrow \begin{cases} \pi_0 = \frac{1}{6} \\ \pi_1 = \frac{1}{3} \\ \pi_2 = \frac{1}{6} \\ \pi_3 = \frac{1}{6} \end{cases}$$

$$\mu_0 = 6 \quad \mu_1 = 3 \quad \mu_2 = 6 \quad \mu_3 = 6$$

$$\begin{bmatrix} 6 & 7 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{cases} \pi_6 + \pi_7 = 1 \\ \pi_6 = \frac{1}{2}\pi_7 \end{cases} \Rightarrow \begin{cases} \pi_6 = \frac{1}{3} \\ \pi_7 = \frac{2}{3} \end{cases}$$

$$\mu_6 = 3 \quad \mu_7 = \frac{3}{2}$$

(3)

$$\lim_{n \rightarrow \infty} P_{ij}^{(n)} = \pi_j = \frac{2}{5}$$

$$h_4 = \lim_{n \rightarrow \infty} P_{44}^{(n)} \quad \lim_{n \rightarrow \infty} P_{41}^{(n)} = 0$$

$$h_5 = 1$$

$$h_4 = \frac{1}{5}h_4 + \frac{1}{5}h_5$$

$$h_4 = \frac{1}{4}$$

(4)

$$\lim_{n \rightarrow \infty} (X_n = 4) = 0$$

$$\lim_{n \rightarrow \infty} (X_n = 5) = 0$$

$$\lim_{n \rightarrow \infty} (X_n = 6) = \frac{1}{18} \quad \frac{1}{2}h_3 + \frac{1}{2}h_4 = \frac{1}{2} \times \frac{1}{9} = \frac{1}{18}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} (X_n = 7) &= \frac{1}{2}h_3 + \frac{1}{2}h_4 \\ &= \frac{1}{2} \times \frac{1}{9} \\ &= \frac{1}{9} \end{aligned}$$

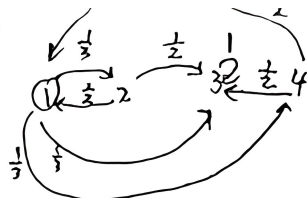
(1)

$$X_n \sim \begin{bmatrix} 0 & 1 \\ 0.5 & 0.5 \end{bmatrix} \quad Y_n \sim \begin{bmatrix} 0 & 1 \\ 0.4 & 0.6 \end{bmatrix}$$

	(0,0)	(0,1)	(1,1)	q
(0,0)	0.2	0.3	0.3	0.2
(0,1)	0.1	0.4	0.4	0.1
(1,0)	0.1	0.4	0.4	0.1
(1,1)	0.2	0.3	0.3	0.2

	(0,0)	(0,1)	(1,1)	(1,0)
(0,0)	1	0	0	0
(0,1)	0.1	0.4	0.4	0.1
(1,1)	0	0	1	0
(1,0)	0.2	0.3	0.3	0.2

23.



$$S_0 = 1$$

(1)

$$\begin{aligned} P(X_1=4, X_3=2) &= P_{1,4} P_{4,2} \\ &= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{2} \\ &= \frac{1}{18} \end{aligned}$$

(2)

$$\begin{aligned} P(X_3=3) &= \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} + \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \\ &= \frac{2}{3} \\ P^2 &= \begin{bmatrix} \frac{1}{3} & 0 & \frac{2}{3} & 0 \\ 0 & \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{6} \end{bmatrix} \end{aligned}$$

(2)

$$P X_n$$

$$\lim_{n \rightarrow \infty} P(\beta_n = (a, d)) = \pi_A$$

$$\pi = (\pi_A, \pi_B, \pi_C, \pi_D)$$

A, B, C, D 代表 0, 1, 1, 0

$$\pi_A + \pi_B + \pi_C + \pi_D = 1$$

$$\lim_{n \rightarrow \infty} P(\beta_n = (0, 0)) = 0 \cdot h_A + h_B + h_C + 0 \cdot h_D = h_B$$

$$\begin{cases} h_A = 1 \\ h_B = 0.1h_A + 0.4h_C + 0.1h_D + 0.4h_B \\ h_C = 0 \\ h_D = 0.2h_D + 0.3h_C + 0.3h_B + 0.2h_A \end{cases}$$

$$\lim_{n \rightarrow \infty} P(\beta_n = (0, 0)) = \frac{2}{9}$$

(3)

$$Q_B^{(A)} = E(T_A < \infty | S_0 = B)$$

$$Q_i^{(N)} = E(T_A < \infty | S_0 = i) \quad i = A, B, C, D$$

$$Q_A^{(A)} = 0$$

$$Q_B^{(A)} = 1 + 0.1Q_A^{(A)} + 0.4Q_B^{(A)} + 0.1Q_D^{(A)}$$

$$Q_D^{(A)} = 1 + 0.2Q_A^{(A)} + 0.3Q_B^{(A)} + 0.2Q_D^{(A)}$$

$$\begin{cases} Q_A = 0 \\ Q_C = 0 \\ Q_B = 0.4Q_B + 0.1Q_D + 1 \\ Q_D = 0.3Q_B + 0.2Q_D + 1 \end{cases} \quad \text{平均 } 2 \frac{1}{3}$$

(3)

$$f_{12}^{(N)} \quad f_{12}^{(0)} = 0 \quad f_{12}^{(1)} = \frac{1}{3} \quad f_{12}^{(2)} = \frac{1}{18}$$

$$f_{12}^{(N)} = \left(\frac{1}{3} \right)^{\frac{N-1}{2}} \times \frac{1}{3} \quad n \text{ 为奇数且 } n \geq 2$$

$$f_{12}^{(N)} = \begin{cases} \frac{1}{3} & n=0 \\ \frac{1}{18} & n=1 \\ \frac{1}{3} \left(\frac{1}{3} \right)^{\frac{n-1}{2}} & n \geq 2 \text{ 且 } n \text{ 为奇数} \end{cases} \quad \text{即 } f_{12}^{(N)} = \begin{cases} 0 & n \text{ 为偶数} \\ \frac{1}{3} \left(\frac{1}{3} \right)^{\frac{n-1}{2}} & n \text{ 为奇数} \end{cases}$$

(4)

$$\begin{cases} h_1 = \frac{1}{3}h_2 + \frac{1}{3}h_4 \\ h_2 = 1 \\ h_3 = 0 \\ h_4 = \frac{1}{2}h_1 \\ h_1 = \frac{2}{5} \end{cases}$$

(5)

$$\begin{cases} a_1 = 1 + \frac{1}{3}a_2 + \frac{1}{3}a_4 \\ a_2 = 1 + \frac{1}{2}a_1 \\ a_3 = 0 \\ a_4 = 1 + \frac{1}{2}a_1 \end{cases}$$

$$\begin{cases} a_1 = \frac{5}{2} \\ a_2 = \frac{9}{4} \\ a_3 = 0 \\ a_4 = \frac{9}{4} \end{cases}$$

$$a_1 = \frac{5}{2} \quad \text{平均 } \frac{5}{2} \text{ 步}$$